

Part I — Foundations of Physics · Chapter 1

What Physics Studies

Introducing the discipline that asks the deepest questions about the physical world

Why does a dropped ball fall rather than rise? Why does a plucked guitar string produce a musical note? Why does rubbing a balloon make it stick to a wall? At its heart, physics is the attempt to answer questions like these — not just to record what happens, but to understand why, and to express that understanding in precise, predictive language.

1.1 What Physics Is

Physics is the most fundamental of the natural sciences. It is the study of matter, energy, motion, forces, space, time, and the relationships among them. Where chemistry asks how atoms combine to form substances, and biology asks how living systems function, physics asks the deeper question beneath both: what are the underlying rules that govern all matter and energy in the universe, and can those rules be stated in precise mathematical language?

The word physics comes from the ancient Greek *physis*, meaning “nature” or “the way things are.” For most of human history, the study of nature was not separated into distinct disciplines. It was Galileo Galilei (1564–1642) and Isaac Newton (1643–1727) who, through careful observation, experiment, and mathematics, demonstrated that the behavior of physical objects follows laws — consistent rules that apply everywhere, to everything, all the time. That insight transformed natural philosophy into the discipline we now call physics.

At its most basic, physics does four things: it observes the physical world carefully; it identifies patterns in those observations; it expresses those patterns as mathematical laws; and it uses those laws to make predictions that can be tested. If the predictions are confirmed by experiment, the law stands. If not, the law is revised or replaced. This cycle of observation, hypothesis, testing, and revision is what makes physics — and science generally — self-correcting and reliable.

“Physics is the attempt to read the book of nature. The universe is written in the language of mathematics.” — Galileo Galilei (paraphrased)

One of the most remarkable features of physics is its extraordinary generality. The

same law of gravitation that explains why an apple falls from a tree also explains the orbits of planets around the Sun and the bending of starlight around massive galaxies. The same laws of electromagnetism that govern the current in a flashlight circuit also explain how radio waves travel across continents and how the human eye detects light. This power of unification — the ability of a small number of laws to explain an enormous range of phenomena — is what makes physics both intellectually beautiful and practically powerful.

Bridge Note *In the MGU Physics advanced series, Volume I revisits Newton's laws using the Lagrangian and Hamiltonian formulations, which reveal even deeper unifying principles beneath the laws you will learn in this book.*

1.2 Why Physics Matters

Physics matters for two distinct reasons: intellectual and practical. The intellectual case is that physics gives us genuine understanding of the natural world. Understanding how a pendulum swings, why the sky is blue, how nuclear reactions power the Sun, or what happens to time near a black hole is intrinsically valuable — it enlarges our sense of what the universe is and where we stand within it.

The practical case is equally compelling. Virtually every technology that defines modern life rests on physics. Electricity, electronics, computers, medical imaging, communications, transportation, energy production, and materials science all emerged from physical understanding. The transistor — the device that made modern computing possible — was invented by physicists applying quantum mechanics to semiconductors. GPS satellites work because their designers accounted for time dilation predicted by Einstein's theory of relativity. MRI machines use the quantum-mechanical spin of atomic nuclei. There is no sharp boundary between "pure" and "applied" physics: today's curiosity-driven research is tomorrow's engineering tool.

Physics also underpins the other sciences. Chemistry is governed by the physics of electrons and atomic bonds. Biology at the molecular level is biochemistry, which is applied chemistry, which is applied quantum physics. Geology depends on fluid dynamics, thermodynamics, and seismology. Atmospheric science is applied thermodynamics and fluid mechanics. Understanding physics is not just useful for physicists: it provides the conceptual foundation for all of science and engineering.

1.3 The Physical World

When physicists refer to the "physical world," they mean everything that exists in space and time and that can be observed or measured. This includes objects of every size, from subatomic particles to galaxy clusters; phenomena of every speed, from nearly

stationary to nearly the speed of light; and energy in every form, from the chemical energy in food to the nuclear energy in stars.

The physical world is organized by scale. Different laws and different tools apply at different scales. At the scale of everyday objects — from a grain of sand to a mountain range — Newton's laws of motion provide an excellent description of reality. At very small scales (the scale of atoms and smaller), quantum mechanics becomes essential. At very high speeds (a substantial fraction of the speed of light), Einstein's special relativity must be used. At very large scales with very strong gravity (the scale of black holes and the universe itself), general relativity applies.

One of the organizing themes of this book is that the same phenomenon can look very different at different scales. Water in a glass looks like a smooth, continuous liquid; viewed at the scale of atoms, it is a collection of discrete molecules in constant, chaotic motion. A current in a wire looks like a smooth flow of electricity; at the atomic level, it is a drift of electrons colliding with metal ions. Keeping both the macroscopic (large-scale) and microscopic (small-scale) pictures in mind simultaneously is one of the most important thinking skills in physics.

Representative Scales of the Physical World

Object or Phenomenon	Approximate Size
Observable universe	$\sim 10^{26}$ m
Milky Way galaxy	$\sim 10^{21}$ m
Distance from Earth to Sun	$\sim 1.5 \times 10^{11}$ m
Earth's radius	$\sim 6.4 \times 10^6$ m
A human	$\sim 1\text{--}2$ m
A grain of sand	$\sim 10^{-3}$ m
A human cell	$\sim 10^{-5}$ m
A bacterium	$\sim 10^{-6}$ m
A water molecule	$\sim 3 \times 10^{-10}$ m
A proton (nucleus)	$\sim 10^{-15}$ m

Table 1.1 *The physical world spans roughly 41 orders of magnitude in size.*

1.4 Matter, Energy, Space, and Time

Four concepts form the foundation of all of physics: matter, energy, space, and time. Everything in the physical world can be described in terms of these four things, and every branch of physics is ultimately concerned with understanding their properties and their relationships.

Matter is anything that has mass and takes up space. It is the “stuff” of the universe: the atoms, molecules, rocks, stars, and living organisms that populate space and time. Matter has properties we can measure: mass, density, temperature, electric charge, and many others. A central goal of physics is to understand how matter behaves under various conditions — how it moves, how it responds to forces, how it stores and transfers energy.

Energy is harder to define than matter because it does not take up space or have a definite location; it is, rather, a property of physical systems — a measure of their ability to cause change. Energy comes in many forms: kinetic energy (energy of motion), potential energy (stored energy due to position or configuration), thermal energy (the energy of random molecular motion), chemical energy, electrical energy, light, and nuclear energy. One of the most important laws in all of physics is the law of conservation of energy: energy cannot be created or destroyed, only converted from one form to another. This law will appear, in different guises, in every part of this book.

Space is the three-dimensional arena in which physical events occur. In everyday life, we describe positions in space using three numbers (think of longitude, latitude, and altitude). In physics, we use coordinate systems — mathematical grids that allow us to measure distances and directions precisely. The properties of space itself — how it curves in the presence of mass — turn out to be intimately connected to gravity, as Einstein showed in his general theory of relativity.

Time is the dimension along which events occur in sequence. Physicists measure time in seconds (or fractions and multiples thereof) and treat it as a fourth dimension, comparable in many ways to the three dimensions of space. One of the most astonishing discoveries of twentieth-century physics is that time is not absolute: it passes at different rates for observers moving at different speeds or in different gravitational fields. This will be explored conceptually in Part IX of this book.

Bridge Note *The advanced volumes treat space and time as a unified four-dimensional ‘spacetime’ described by the metric tensor — a concept that begins to make sense only after mastering vectors and coordinate systems. The intuition built here will make that step far less daunting.*

1.5 Motion, Forces, and Interactions

If matter and energy are the “what” of physics, motion, forces, and interactions are the “how.” Motion is the change of position over time. Forces are the agents that cause changes in motion. Interactions are the ways in which objects affect one another through forces.

The study of motion without reference to its causes is called kinematics. It answers questions like: How fast is an object moving? In what direction? How is its velocity changing? Kinematics uses mathematics — algebra, geometry, and graphs — to describe motion precisely. In Part II of this book you will develop a complete kinematic toolkit for objects moving in straight lines and in two dimensions.

The study of the causes of motion is called dynamics. Here, Isaac Newton’s three laws of motion are the foundational tools. Newton’s first law says that an object at rest stays at rest, and an object in motion keeps moving in a straight line at constant speed, unless acted upon by a net force. Newton’s second law quantifies how a force changes motion: force equals mass times acceleration. Newton’s third law says that forces always come in pairs: whenever object A exerts a force on object B, object B exerts an equal and opposite force on object A.

In nature, there are four fundamental interactions, or fundamental forces: gravity, electromagnetism, the strong nuclear force, and the weak nuclear force. Gravity and electromagnetism operate over large distances and are the forces of everyday experience. The strong and weak nuclear forces operate only at the subatomic scale and are responsible for holding atomic nuclei together and for radioactive decay, respectively. Most of the physics in this book — motion, machines, heat, sound, electricity, light — involves gravity and electromagnetism.

1.6 Systems and Models

One of the most important thinking tools in physics is the concept of a system. A system is any collection of objects, particles, or regions of space that we choose to study as a unit. Everything outside the system is called the environment or surroundings. The choice of system is the physicist’s first and most important decision in analyzing any problem, because it determines what forces and interactions need to be considered.

For example, if we want to analyze how a soccer ball moves after being kicked, we might choose the ball as our system. The force of the kick, air resistance, and gravity are then all external forces acting on that system. If instead we choose the ball-and-Earth as our combined system, then gravity becomes an internal force and is automatically accounted for by conservation of momentum.

Models are simplified representations of physical systems. No physical situation is so simple that it can be described exactly and completely by equations — there are always complications: air resistance, friction, small asymmetries, tiny temperature variations. A model deliberately ignores some of these complications in order to focus on the essential physics. A frictionless surface, a point mass, an ideal spring, an ideal gas: these are all models. The art of physics is knowing which complications can safely be ignored for the question at hand, and which cannot.

Throughout this book you will encounter many models. You will treat the Earth as a uniform sphere when computing gravitational force; you will treat gases as collections of tiny, perfectly elastic balls; you will treat light as a ray for studying lenses and as a wave for studying interference. Each model is an approximation, but each is also powerful and reliable within its domain.

1.7 Observation, Measurement, and Evidence

Physics is an empirical science: its claims must ultimately be grounded in observation and measurement, not in authority, tradition, or pure reasoning alone. An observation is a recorded sensory experience of the physical world. A measurement is a quantitative observation — one that assigns a numerical value and a unit to some physical quantity.

Measurement is not passive. Every measurement involves choosing an instrument, reading a scale, recording a result, and estimating uncertainty. No measurement is perfectly precise. The length of a table measured with a ruler accurate to 1 mm cannot be stated to 0.001 mm. This is not a failure of physics; it is a fundamental feature of empirical science. Quantifying and honestly reporting uncertainty is as important as reporting the measurement itself.

Evidence in physics is never a single measurement. It is a pattern of consistent measurements, repeated under varied conditions, by multiple investigators, using multiple methods, that all point to the same conclusion. A result obtained once, in one laboratory, is interesting. A result confirmed by dozens of independent experiments, using different techniques, is knowledge.

The relationship between theory and evidence is subtle. A theory is not merely a guess; in science, it is a well-developed, internally consistent framework that explains a large body of evidence and makes specific, testable predictions. Theories are not “proved” — no finite set of observations can logically prove a universal claim. But they can be disproved — a single well-established observation that contradicts a theory’s prediction is enough to require the theory to be revised or abandoned.

1.8 Laws, Theories, and Explanations

Students sometimes confuse the everyday and scientific meanings of common words. In everyday language, a “theory” is just a guess. In science, it is the highest level of understanding — a deeply tested, broadly applicable, predictively powerful framework. Similarly, a “law” in everyday language means a rule imposed by authority. In physics, a law is an observed regularity in nature that has been confirmed so thoroughly, and that holds so generally, that we treat it as a fixed feature of the universe.

The distinction between a law and a theory is sometimes misunderstood. Newton’s second law ($F = ma$) is a law: a mathematical statement of how force and acceleration are related. Newton’s theory of gravitation is a theory: a framework that explains not just how things fall, but why they orbit, how tides arise, and how the Moon and Earth interact. The law tells you what happens; the theory explains why, and predicts what will happen in new situations.

A scientific explanation is an account of a phenomenon in terms of more fundamental principles. When we explain why a ball accelerates when you push it by saying “because of Newton’s second law,” we are explaining a specific event in terms of a general principle. When we explain Newton’s second law in terms of the underlying structure of spacetime and the nature of mass, we are going one level deeper. Physics is a hierarchy of explanations, and the search for deeper levels is one of its driving forces.

Term	What It Means in Science
Hypothesis	A proposed explanation to be tested; not yet confirmed by evidence
Law	A well-confirmed mathematical regularity observed in nature
Theory	A broad, tested framework explaining many phenomena and making predictions
Model	A simplified representation focusing on essential features
Experiment	A controlled test designed to yield evidence about a specific question
Uncertainty	The range of values within which a measurement is

	reliable
--	----------

Table 1.2 *Scientific vocabulary often differs from everyday usage.*

1.9 Major Branches of Physics

Physics is not a single, monolithic subject. Over centuries it has differentiated into specialized branches, each with its own set of tools, phenomena, and questions. The branches are not isolated, however; they constantly borrow from and inform one another.

Branch	What It Studies
Classical mechanics	Motion, forces, energy, and momentum of objects at everyday speeds and sizes
Thermodynamics	Heat, temperature, energy, entropy, and the laws governing thermal processes
Fluid mechanics	The behavior of liquids and gases at rest and in motion
Wave physics & acoustics	Oscillations, mechanical waves, sound, and their interactions with matter
Optics	Light: its propagation, reflection, refraction, diffraction, and polarization
Electromagnetism	Electric and magnetic fields, charges, currents, and electromagnetic waves
Quantum mechanics	The behavior of matter and energy at the scale of atoms and subatomic particles
Special relativity	The physics of objects moving at speeds close to the speed of light
General relativity	Gravity as the curvature of spacetime; black holes and cosmology
Nuclear physics	The structure and reactions of atomic nuclei; fission, fusion, radioactivity

Particle physics	The fundamental constituents of matter and the forces between them
Condensed matter	The properties of solids, liquids, and other phases of matter
Astrophysics & cosmology	Stars, galaxies, the large-scale structure and origin of the universe
Biophysics	Physical principles in biological systems: mechanics, heat, electricity in life

Table 1.3 *Major branches of physics and their subject matter.*

This book follows a logical progression through most of these branches: mechanics first, then waves, heat, fluids, electricity and magnetism, and finally an introduction to modern physics. Part X surveys applied branches. The MGU Physics advanced five-volume series covers all of these at the university level, building on every concept introduced here.

1.10 Physics in Everyday Life

Physics is not something that happens only in laboratories. Every moment of ordinary life is saturated with physical phenomena. When you walk, Newton's laws govern the forces between your feet and the ground. When you throw a ball, kinematics describes the arc it follows. When you boil water, thermodynamics describes the energy transfer from stove to pot to liquid. When you listen to music through headphones, wave physics and electromagnetism describe every step from the electrical signal to the sound waves in your ear.

Consider a simple act: turning on a lamp. This involves electric charge flowing through a conductor, a current limited by the resistance of the filament or LED, the conversion of electrical energy to light (electromagnetic radiation), and the propagation of that light through the room. Quantum mechanics explains exactly why an LED emits the color it does. The circuit breaker that protects the wiring relies on electromagnetism and thermal expansion. The transformer on the power pole outside converts voltage using Faraday's law of induction. The entire act, from flipping a switch to seeing light, is a cascade of physical phenomena.

One of the goals of this book is to give you the conceptual vocabulary to recognize physical phenomena when you see them in daily life — and to ask the right questions about them. Why does a spinning top stay upright? Why does a wet road reflect headlights more than a dry road? Why does your voice echo in a large empty room?

These are not trick questions; they have clear physical answers, and by the end of this book you will be equipped to begin finding them.

1.11 Physics, Technology, and Society

The relationship between physics and technology is sometimes described as “physics enables technology.” This is true but incomplete. Technology also drives physics: the need to build better instruments has repeatedly forced physicists to develop new theories. The telescope revealed the structure of the solar system. The steam engine motivated the development of thermodynamics. The effort to understand the properties of semiconductors led to quantum mechanics being applied to solid-state physics, which led to the transistor, the microchip, and the digital age.

Physics also has profound social consequences. Nuclear physics gave the world both nuclear power plants and nuclear weapons. Our understanding of the greenhouse effect, which is atmospheric physics, underpins the science of climate change. The physics of solar cells and wind turbines is central to the global energy transition. Decisions about these technologies — how to use them, how to regulate them, how to share their benefits and manage their risks — are among the most important decisions that societies must make in the twenty-first century.

For this reason, physics education is not only about preparing future scientists and engineers. It is about equipping citizens with the conceptual tools to evaluate scientific claims, understand technological options, and participate meaningfully in public decisions that have a physical dimension. The ability to reason about energy, forces, radiation, and uncertainty is genuinely useful for everyone — not just those who will spend their careers in a laboratory.

1.12 How to Think Like a Physics Student

Physics has a distinctive way of approaching problems, and learning that approach is itself one of the most valuable things you will gain from this course. Here are the habits of mind that separate effective physics students from those who struggle.

Draw before you calculate. A diagram — a picture of the situation, a free-body diagram showing forces, a graph of motion — almost always makes the physical situation clearer and the mathematics easier. Professional physicists draw constantly. Students who try to go straight to equations almost always make more errors than those who diagram first.

Define your variables and write your equations symbolically before substituting numbers. Algebra done with symbols is easier to check and easier to understand than

arithmetic done with numbers. It also lets you see how the answer depends on the quantities involved — a powerful check on whether your equation makes sense.

Always include units. A number without units is not a physical quantity; it is just a number. The units must be carried through every calculation and must come out correctly at the end. If you compute a speed and get an answer in units of $\text{kg}\cdot\text{m}$, something has gone wrong. Units are a continuous check on your work.

Estimate first, calculate second. Before doing a detailed calculation, estimate the answer. Is it reasonable? If you calculate that a ball thrown upward takes 1000 seconds to reach its peak, something has gone wrong. Developing physical intuition — a feel for reasonable magnitudes — is one of the most important skills you can build.

Ask what would change. After solving a problem, vary one of the given quantities in your mind and ask how the answer would change. If the mass of an object increases, does the acceleration increase or decrease? If the angle of a slope doubles, what happens to the component of gravity along the slope? This kind of “what if” reasoning builds deep understanding faster than memorizing formulas.

Embrace confusion as a starting point. Physics problems are often confusing at first. That is not a sign that you lack ability; it is a sign that the problem is doing its job. Sit with the confusion. Read the problem again. Draw the diagram. Write down what you know and what you are looking for. The path forward almost always reveals itself to students who are patient and systematic.

Bridge Note *The thinking habits in this section — especially dimensional analysis, symbolic algebra, and estimation — are not just useful here. They are the entry point to the calculus, linear algebra, and tensor methods used throughout the MGU Physics advanced series.*

Chapter Summary

Physics is the study of matter, energy, space, time, motion, forces, and their relationships. It is the most fundamental natural science, providing the conceptual foundation for all other sciences and for modern technology.

Key ideas from Chapter 1:

- Physics identifies patterns in the natural world and expresses them as mathematical laws that allow precise prediction.
- The four fundamental concepts of physics are matter, energy, space, and time.

- Motion is described by kinematics; the causes of motion are explained by dynamics, using Newton's laws of force.
- A system is any defined collection of objects; a model is a deliberate simplification that focuses on essential physics.
- Measurement is the foundation of physics; all measurements have uncertainty, which must be estimated and reported.
- A scientific law is a well-confirmed mathematical regularity; a scientific theory is a broad explanatory framework.
- Physics branches include classical mechanics, thermodynamics, electromagnetism, optics, quantum mechanics, and relativity, among others.
- Physics enables technology and is central to major societal decisions about energy, environment, and medicine.
- Effective physics thinking involves drawing diagrams, working symbolically, checking units, estimating, and embracing confusion as a starting point.

Key Terms

physics: The study of matter, energy, space, time, motion, forces, and their relationships; the most fundamental natural science.

matter: Anything that has mass and occupies space.

energy: A property of physical systems measuring their capacity to cause change; exists in many forms and is conserved.

space: The three-dimensional arena in which physical events occur.

time: The dimension along which events occur in sequence; not absolute but relative to the observer's motion and gravitational environment.

kinematics: The description of motion without reference to its causes.

dynamics: The study of the forces that cause changes in motion.

force: An interaction that can change the motion of an object; a vector quantity measured in newtons.

system: A defined collection of objects or regions chosen for study; everything outside the system is the environment.

model: A deliberate simplification of a physical situation that retains the essential physics while ignoring secondary complications.

observation: A recorded sensory experience of the physical world.

measurement: A quantitative observation assigning a numerical value and a unit to a physical quantity.

uncertainty: The range of values within which a measurement is reliable; inherent in all empirical observations.

scientific law: A well-confirmed, broadly applicable mathematical statement of a regularity in nature.

scientific theory: A broadly applicable, well-tested explanatory framework that makes specific, testable predictions.

hypothesis: A proposed, testable explanation that has not yet been confirmed by sufficient evidence.

macroscopic: At the scale of everyday objects, visible to the unaided eye.

microscopic: At the scale of atoms, molecules, or smaller structures, invisible to the unaided eye.

fundamental forces: The four basic interactions in nature: gravity, electromagnetism, the strong nuclear force, and the weak nuclear force.

Concept Review Questions

Answer the following in complete sentences. No formulas are required.

1. In your own words, what does it mean to say that physics is the most 'fundamental' of the natural sciences? Give one example of how physics underlies another science.
2. What is the difference between an observation and a measurement? Why is measurement important in physics?
3. Describe the difference between a scientific law and a scientific theory. Use Newton's second law and Newton's theory of gravitation as examples.
4. What is a model in physics? Give one example of a physical model used in everyday problem-solving and explain what it ignores.
5. Name the four fundamental forces of nature. Which two are most relevant to everyday physics? Which two operate only at the subatomic scale?
6. What is a system in physics? Why is the choice of system important when analyzing a physical problem?
7. Explain, in your own words, why no measurement can be perfectly precise. What does it mean to report the uncertainty of a measurement?

8. Physics is described as “self-correcting.” What does this mean? How does evidence lead to the revision or replacement of physical laws and theories?
 9. Give two examples of physical phenomena from everyday life. For each one, identify which branch of physics is most relevant.
 10. Describe three habits of mind that distinguish effective physics students from those who struggle. For each one, explain why it helps.
-

Short Answer Questions

These questions require slightly longer written responses or simple reasoning. No calculation is needed.

11. A student claims: “The theory of evolution is ‘just’ a theory, so it’s not really proven.” How would a physicist respond to this claim? What does ‘theory’ mean in science?
 12. The same law of gravitation explains falling apples and the orbits of planets. What does this tell us about the nature of physical laws?
 13. Physics involves both macroscopic and microscopic descriptions of matter. Describe what water looks like at each scale, and explain why both descriptions are useful.
 14. Give an example of a technological development that required new physical understanding, and an example of a technological need that drove new physics. What does this tell you about the relationship between physics and technology?
 15. A classmate says: “I don’t need to study physics; I’m going to be a doctor.” Construct a response that uses specific examples from medicine or biology to argue that physical understanding is relevant to a career in medicine.
 16. Name all four fundamental forces. For each one, give one example of where it plays a dominant role in nature.
 17. Explain why a physicist might deliberately build a model that ignores friction, even when studying a situation where friction is clearly present. Under what circumstances would such a model be acceptable? When would it fail?
 18. What is the difference between kinematics and dynamics? Give a physical situation and explain how each would describe it differently.
-

Physics in Everyday Life Activity

Physics Observatory Journal — One Week Assignment

For five consecutive days, keep a brief daily log. Each day, identify one physical phenomenon you observe in your normal life — something you see, hear, feel, or experience. For each observation, record the following:

- a) What you observed (describe it in one or two sentences).
19. What branch of physics you think is most relevant (refer to Table 1.3 if needed).
20. One question you have about the physics involved.
21. Your best guess at an explanation, even if you are not sure.

At the end of the week, review your five entries and answer: Were there any patterns? Did the same branch of physics come up more than once? Did any of your guesses at explanations change as you thought about them more?

Your log does not need to be long. Two to four sentences per day is sufficient. The goal is to begin developing the habit of seeing physics in the world around you.

Chapter 1 Checkpoint

Before moving on to Chapter 2, you should be able to do the following without looking at your notes. If you cannot, review the relevant section.

I can...	See Section
Define physics and state what it studies	1.1
Give at least two reasons why physics matters (one intellectual, one practical)	1.2
Explain what macroscopic and microscopic mean and give an example of each	1.3
Define matter, energy, space, and time in physical terms	1.4
Distinguish kinematics from dynamics	1.5

Explain what a system is and why choosing a system matters	1.6
State the difference between an observation and a measurement	1.7
Distinguish a scientific law from a scientific theory	1.8
Name five or more branches of physics and their subject matter	1.9
Identify a physical phenomenon in everyday life and name its relevant branch	1.10
Give one example of physics enabling technology and one of technology driving physics	1.11
List at least four problem-solving habits of effective physics students	1.12

Continue to Chapter 2: Measurement, Units, and Scientific Notation →